

Makar Glazachev

BACKEND, DATA & ML ENGINEER

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Works across the whole of a data product — the pipeline underneath, the service on top, the model in between — and ships it tested, containerised and documented.

EXPERIENCE

Transmashholding (TMH) — Data Engineering Intern Moscow · Jul 2026

Russia's largest rail rolling-stock manufacturer; joined after winning its hackathon.

- Built a **4,400-line ETL pipeline** turning raw Excel exports — locomotive movements, depot equipment, station coordinates — into **16 analysis-ready Parquet tables** on a versioned, documented schema (DuckDB, PyArrow, RapidFuzz).
- Modelled **30,146 movement events**, **20,866 locomotives** and 369 enterprises, and built a **209,638-row search index** resolving locomotive numbers with or without leading zeros and down to individual sections.
- Shipped an interactive map visualiser: stations plotted and filterable by railway, visit frequency per period, locomotives traced over time.
- Backed it with 12 pytest tests, ruff gates, Docker Compose packaging and written data-model and validation docs.

HACKATHONS

3 entered · 3 won

Yandex Vibe Coding Hackathon — Winner — captain & sole developer Apr 2026

Platform taking a project proposal from intake to a public showcase, run by **16 agents** on YandexGPT: a review panel of four personas scoring in parallel and a moderator issuing a weighted 0–100 verdict, a **nine-agent** deep-research chain through architecture, roadmap, staffing and risk, and two MCP agents turning the result into Yandex Tracker and SourceCraft work.

TMH Hackathon, Sirius — Winner — ML engineer, team Ascent Feb 2026

Wheel-life track, one of three contested by **14 teams**. Owned the prediction model: CatBoostRegressor over **407,669 wheel-wear records** in two variants — baseline hypotheses and the team's full feature set — blended with fitted weights, as per-decile error showed neither held the full range. **MSE 0.12**, served over a REST API. github.com/Astal2307/TMHackathon_Ascent

Young Scientists Hackathon, Sirius — Winner, international field Nov 2025

Academic Profile — publication records harvested from Scopus, Web of Science, Google Scholar and Crossref, turned into h-index and citation metrics, a knowledge graph and collaborator recommendations. **Sole backend engineer** — FastAPI, 15 REST endpoints, JWT auth and SQLAlchemy models, serving the team's TF-IDF + KNN recommender over **194,849 researchers**. github.com/brainstorm-sirius/academic-profile

SELECTED PROJECTS

Chemical Feedstock Selection System — built with SIBUR experts Python · Jul 2025

From warehouse stock and a reaction graph — substance transitions under given operations — the system enumerates **every** route to a target compound by solving the corresponding linear systems, then ranks them by difficulty.

Score Editor — sheet music editor C++17 · Qt6 · Mar 2026

Desktop score editor — notes, rests, accidentals, barlines and multi-staff layout under strict positioning rules, PNG/PDF export, JSON projects. Live pitch input: a **hand-implemented YIN detector** places each note played into the right staff position. github.com/MakarON-228/Note_redactor

EDUCATION

Sirius University of Science and Technology 2024 – 2029

Design, Development and Management of Complex Information Systems — five-year integrated programme

- Russia's first degree to combine secondary and higher education, admitting students straight after 9th grade, under an experimental legal regime in the Sirius Federal Territory.
- Core track: software architecture for critical infrastructure, systems analysis, enterprise systems, process automation.
- Industry projects from year one with Yandex, VK, Rosatom and Russian Railways; the final year is a full placement.

TECHNICAL SKILLS

LANGUAGES	Python, C++17, SQL
BACKEND	FastAPI, SQLAlchemy, Pydantic, REST API design, JWT authentication
DATA	DuckDB, Parquet / PyArrow, pandas, NumPy, SciPy, RapidFuzz, ETL design
ML & AI	CatBoost, scikit-learn, model serving, multi-agent LLM systems, MCP
TOOLING	Docker & Compose, Git, pytest, ruff, Qt6, CMake